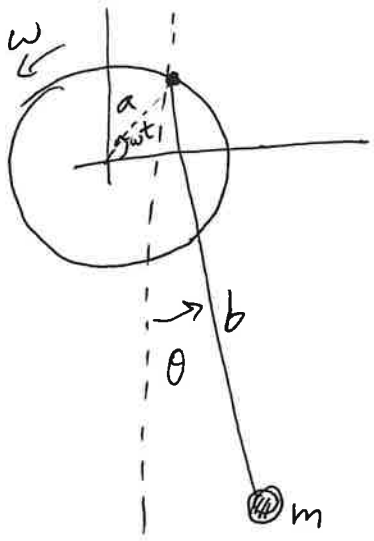
PROBLEM SET #4 (SOLUTIONS)

1)



a) (i) $x = a\cos\omega t + b\sin\theta \qquad y = a\sin\omega t - b\cos\theta$

(ii) $\dot{x} = -a\omega\sin\omega t + b\dot{\theta}\cos\theta \qquad \dot{y} = a\omega\cos\omega t + b\dot{\theta}\sin\theta$

(iii) $\ddot{x} = -a\omega^2\cos\omega t + b(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta)$

$\ddot{y} = -a\omega^2\sin\omega t + b(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta)$

b) $L = T - V, \qquad T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2), \qquad V = mgy$

$$L = \frac{1}{2}m\left(a^2\omega^2 + b^2\dot{\theta}^2 - 2ab\omega\dot{\theta}\sin(\omega t)\cos\theta + 2ab\omega\dot{\theta}\cos(\omega t)\sin\theta\right) - mg(a\sin(\omega t) - b\cos(\theta))$$

$2ab\omega\dot{\theta}[\cos\omega t\sin\theta - \sin\omega t\cos\theta] = \sin(\theta - \omega t)$

$$\Rightarrow \boxed{L(\theta,\dot{\theta}) = \frac{1}{2}m\left[a^2\omega^2 + b^2\dot{\theta}^2 + 2b\dot{\theta}a\omega\sin(\theta - \omega t)\right] - mg(a\sin\omega t - b\cos\theta)}$$

1) Continued...

$$\theta: \frac{d}{dt}\frac{\partial L}{\partial \dot\theta} = mb^2\ddot\theta + mba\omega(\dot\theta-\omega)\cos(\theta-\omega t)$$

$$\frac{\partial L}{\partial \theta} = mb\dot\theta a\omega\cos(\theta-\omega t) - mgb\sin\theta$$

$$\Rightarrow \frac{d}{dt}\frac{\partial L}{\partial \dot\theta} - \frac{\partial L}{\partial \theta} = 0$$

$$mb^2\ddot\theta + mba\omega(\dot\theta-\omega)\cos(\theta-\omega t) = mb\dot\theta a\omega\cos(\theta-\omega t) - mgb\sin\theta$$

$$\ddot\theta = \frac{mb\dot\theta a\omega}{mb^2}\cos(\theta-\omega t) - \frac{mgb\sin\theta}{mb^2} - \frac{mba\omega(\dot\theta-\omega)}{mb^2}\cos(\theta-\omega t)$$

$$= \frac{mba\omega^2}{mb^2}\cos(\theta-\omega t) - \frac{mgb\sin\theta}{mb^2}$$

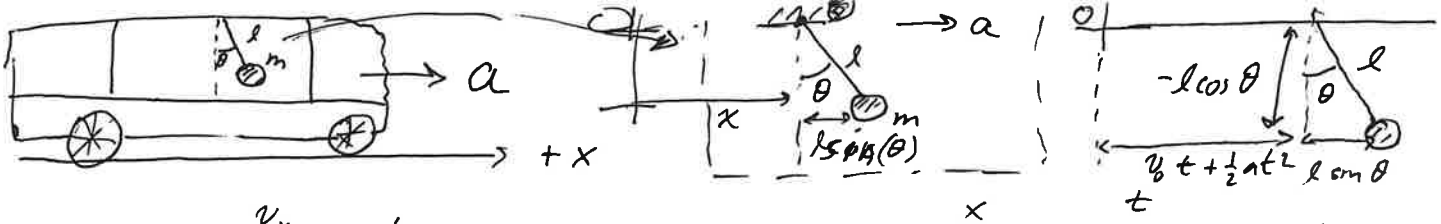
$$\boxed{\ddot\theta = \frac{\omega^2 a}{b}\cos(\theta-\omega t) - \frac{g}{b}\sin\theta}$$

For $\omega \to 0$, $\ddot\theta = \frac{\omega^2 a}{b}\cos(\theta) - \frac{g}{b}\sin\theta$

$$\boxed{\ddot\theta \simeq -\frac{g}{b}\theta} \quad \theta \approx 0 \to \sin\theta \approx \theta$$

Stationary pendulum equation

2)



a) $a = \frac{dv_x}{dt}$ $\int_{v_0}^{v_x} dv_x = \int_0^t a\,dt \rightarrow v_x = v_0 + at$ $\int_0^x dx = \int_0^t v_0\,dt + \int_0^t at\,dt$

$x = v_0 t + \frac{1}{2}at^2$

pendulum coordinates

$x = v_0 t + \frac{1}{2}at^2 + l\sin\theta$

$y = -l\cos\theta$

pendulum velocity

$\dot{x} = v_0 + at + l\dot{\theta}\cos\theta$

$\dot{y} = l\dot{\theta}\sin\theta$

b) $L = T - V$, $T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2)$, $V = mgy$

$\Rightarrow T = \frac{1}{2}m\left[(v_0 + at) + l\dot{\theta}\cos\theta\right)^2 + (l\dot{\theta}\sin\theta)^2\right]$

$= \frac{1}{2}m\left[(v_0+at)^2 + 2(v_0+at)l\dot{\theta}\cos\theta + \underbrace{l^2\dot{\theta}^2\cos^2\theta + l^2\dot{\theta}^2\sin^2\theta}_{l^2\dot{\theta}^2}\right]$

$V = -mgl\cos\theta$

$$\boxed{L = \frac{1}{2}m\left[(v_0+at)^2 + 2(v_0+at)l\dot{\theta}\cos\theta + l^2\dot{\theta}^2\right] + mgl\cos\theta}$$

$\hookrightarrow L(\theta, \dot{\theta})$

θ: $\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0$, $\frac{\partial L}{\partial \dot{\theta}} = \frac{1}{2}m\left[2(v_0+at)l\cos\theta + 2l^2\dot{\theta}\right]$

$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} = ma\,l\cos\theta - m(v_0+at)l\dot{\theta}\sin\theta + ml^2\ddot{\theta}$

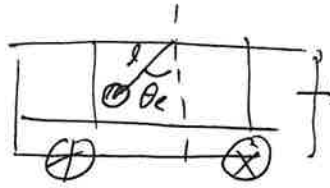
$\frac{\partial L}{\partial \theta} = \frac{1}{2}m\left[-2(v_0+at)l\dot{\theta}\sin\theta\right] - mgl\sin\theta$

$\Rightarrow ma\,l\cos\theta - m(v_0+at)l\dot{\theta}\sin\theta + l^2\ddot{\theta}m + m(v_0+at)l\dot{\theta}\sin\theta + mgl\sin\theta = 0$

$l^2\ddot{\theta} = -a\,l\cos\theta - gl\sin\theta$

2) Continued...

$$\ddot{\theta} = -\frac{g}{\ell}\sin\theta - \frac{a}{\ell}\cos\theta$$

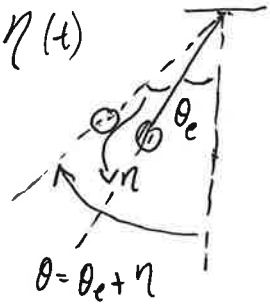
c)  → a at equilibrium, $\ddot{\theta} = 0$ (no angular acceleration)

$$\Rightarrow 0 = -\frac{g}{\ell}\sin\theta_e - \frac{a}{\ell}\cos\theta_e$$

$$\Rightarrow \frac{g\sin\theta_e}{\cos\theta_e} + a\frac{\cos\theta_e}{\cos\theta_e} = \frac{0}{\cos\theta_e} \Rightarrow \tan\theta_e = -\frac{a}{g}$$

$\lim a \to 0$ clearly, $\tan\theta_e \to 0$ ($\theta_e \to 0$), as expected.

d) for small angle displacement from equil. $\theta = \theta_e + \eta(t)$



$$\Rightarrow \ddot{\theta} = \ddot{\eta} = -\frac{g}{\ell}\sin(\theta_e+\eta) - \frac{a}{\ell}\cos(\theta_e+\eta)$$

$$\ddot{\eta} = -\frac{g}{\ell}(\sin\theta_e \underbrace{\cos\eta}_{\approx 1} + \sin\eta\cos\theta_e) - \frac{a}{\ell}(\cos\theta_e \underbrace{\cos\eta}_{\approx 1} - \sin\theta_e \underbrace{\sin\eta}_{\approx \eta})$$

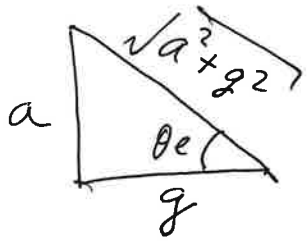
$$= -\frac{g}{\ell}(\sin\theta_e + \eta\cos\theta_e) - \frac{a}{\ell}(\cos\theta_e - \eta\sin\theta_e)$$

$$= -\frac{1}{\ell}\left[\underbrace{(g\sin\theta_e + a\cos\theta_e)}_{=0} + \eta(g\cos\theta_e - a\sin\theta_e)\right]$$

Recall: $\tan\theta_e = \frac{\sin\theta_e}{\cos\theta_e} = -\frac{a}{g} \Rightarrow g\sin\theta_e = -a\cos\theta_e$

2) Continued...

d) $\ddot{\eta} = -\frac{1}{\ell}(g\cos\theta_e - a\sin\theta_e)\eta$



$$\tan\theta_e = -\frac{a}{g}$$

$$\Rightarrow \cos\theta_e = \frac{g}{\sqrt{a^2+g^2}}, \quad \sin\theta_e = \frac{-a}{\sqrt{a^2+g^2}}$$

→ one of these two needs to have a "−" sign (since $\tan\theta_e$ has "−")

$$\Rightarrow \ddot{\eta} = -\frac{1}{\ell}\left(g\cdot\frac{g}{\sqrt{a^2+g^2}} + a\cdot\frac{a}{\sqrt{a^2+g^2}}\right)\eta$$

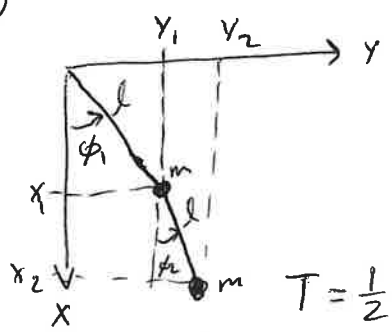
$$\frac{g^2+a^2}{(g^2+a^2)^{1/2}} \to \sqrt{a^2+g^2}$$

$$\Rightarrow \boxed{\ddot{\eta} = -\frac{\sqrt{a^2+g^2}}{\ell}\eta}, \quad \text{where } \omega^2 \equiv \frac{\sqrt{a^2+g^2}}{\ell}$$

In the limit $a \to 0$ (tram stops)

$\Rightarrow \omega^2 = \frac{g}{\ell}$ (pendulum frequency as expected)

3)



$$x_1 = l\cos\phi_1 \qquad \dot{x}_1 = -l\dot{\phi}_1\sin\phi_1$$
$$y_1 = l\sin\phi_1 \qquad \dot{y}_1 = l\dot{\phi}_1\cos\phi_1$$
$$x_2 = l\cos\phi_1 + l\cos\phi_2 \qquad \dot{x}_2 = -l\dot{\phi}_1\sin\phi_1 - l\dot{\phi}_2\sin\phi_2$$
$$y_2 = l\sin\phi_1 + l\sin\phi_2 \qquad \dot{y}_2 = l\dot{\phi}_1\cos\phi_1 + l\dot{\phi}_2\cos\phi_2$$

$$T = \frac{1}{2}m(\underbrace{\dot{x}_1^2 + \dot{y}_1^2}_{l^2\dot{\phi}_1^2}) + \frac{1}{2}m(\underbrace{\dot{x}_2^2 + \dot{y}_2^2}_{l^2\dot{\phi}_1^2 + l^2\dot{\phi}_2^2 + 2l^2\dot{\phi}_1\dot{\phi}_2\sin\phi_1\sin\phi_2 + 2l^2\dot{\phi}_1\dot{\phi}_2\cos\phi_1\cos\phi_2})$$

$$\Rightarrow T = \frac{m}{2}l^2\left[2\dot{\phi}_1^2 + \dot{\phi}_2^2 + 2\dot{\phi}_1\dot{\phi}_2\cos(\phi_1-\phi_2)\right] \qquad 2l^2\dot{\phi}_1\dot{\phi}_2\cos(\phi_1-\phi_2)$$

$$\boxed{T = \frac{m}{2}l^2\left[2\dot{\phi}_1^2 + \dot{\phi}_2^2 + 2\dot{\phi}_1\dot{\phi}_2\cos(\phi_1-\phi_2)\right]}$$

$$V = -mgx_1 - mg(x_2) = -mg(x_1+x_2)$$

$$\Rightarrow \boxed{V = -mgl\left[2\cos\phi_1 + \cos\phi_2\right]}$$

$$L = T - V \Rightarrow \boxed{L = ml^2\left[\dot{\phi}_1^2 + \frac{1}{2}\dot{\phi}_2^2 + \dot{\phi}_1\dot{\phi}_2\cos(\phi_1-\phi_2)\right] + mgl\left[2\cos\phi_1 + \cos\phi_2\right]}$$

$$L(\phi_1, \dot{\phi}_1, \phi_2, \dot{\phi}_2)$$

$$\phi_1: \quad \frac{\partial L}{\partial \phi_1} = ml^2\dot{\phi}_1\dot{\phi}_2\sin(\phi_1-\phi_2) - 2mgl\sin\phi_1$$

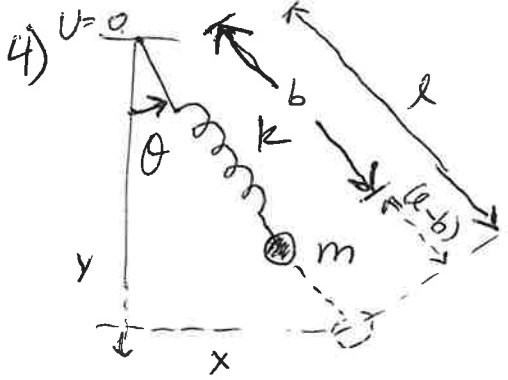
$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\phi}_1} = \frac{d}{dt}\left[2ml^2\dot{\phi}_1 + ml^2\dot{\phi}_2\cos(\phi_1-\phi_2)\right]$$
$$= 2ml^2\ddot{\phi}_1 + ml^2\ddot{\phi}_2\cos(\phi_1-\phi_2) + ml^2\dot{\phi}_2^2\dot{\phi}_1\sin(\phi_1-\phi_2)$$

$$\phi_2: \quad \frac{\partial L}{\partial \phi_2} = -ml^2\dot{\phi}_1\dot{\phi}_2\sin(\phi_1-\phi_2) - mgl\sin\phi_2$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\phi}_2} = \frac{d}{dt}\left[ml^2\dot{\phi}_2 + ml^2\dot{\phi}_1\cos(\phi_1-\phi_2)\right]$$
$$= ml^2\ddot{\phi}_2 + ml^2\ddot{\phi}_1\cos(\phi_1-\phi_2) + ml^2\dot{\phi}_1\dot{\phi}_1\dot{\phi}_2\sin(\phi_1-\phi_2)$$

$$\Rightarrow \phi_1: \frac{d}{dt}\frac{\partial L}{\partial \dot{\phi}_1} - \frac{\partial L}{\partial \phi_1} = 0 \Rightarrow \boxed{2\ddot{\phi}_1 + \ddot{\phi}_2\cos(\phi_1-\phi_2) + \dot{\phi}_2^2\sin(\phi_1-\phi_2) + 2\frac{g}{l}\sin\phi_1 = 0}$$

$$\Rightarrow \phi_2: \frac{d}{dt}\frac{\partial L}{\partial \dot{\phi}_2} - \frac{\partial L}{\partial \phi_2} = 0 \Rightarrow \boxed{\ddot{\phi}_2 + \ddot{\phi}_1\cos(\phi_1-\phi_2) - \dot{\phi}_1^2\sin(\phi_1-\phi_2) + \frac{g}{l}\sin\phi_2 = 0}$$

4) 

$b \rightarrow$ un extended length of spring

$l \rightarrow$ variable length of spring

$x = l\sin\theta \qquad y = l\cos\theta$

$\dot{x} = \dot{l}\sin\theta + l\dot{\theta}\cos\theta$

$\dot{y} = \dot{l}\cos\theta - l\dot{\theta}\sin\theta$

$$T = \frac{1}{2}m(\dot{x}^2+\dot{y}^2) = \frac{1}{2}m(\dot{l}^2 + l^2\dot{\theta}^2)$$

$$U = \frac{1}{2}k(l-b)^2 - mgl\cos\theta$$

$$\boxed{L = T-U = \frac{1}{2}m(\dot{l}^2+l^2\dot{\theta}^2) - \frac{1}{2}k(l-b)^2 + mgl\cos\theta} \quad L(l,\dot{l},\theta,\dot{\theta})$$

$$l: \frac{\partial L}{\partial l} = \frac{1}{2}m(2l\dot{\theta}^2) - \frac{1}{2}k2(l-b) + mg\cos\theta$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{l}} = \frac{d}{dt}\left[\frac{1}{2}m(2\dot{l})\right] = m\ddot{l}$$

$$\theta: \frac{\partial L}{\partial \theta} = -mgl\sin\theta$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} = \frac{d}{dt}\left[\frac{1}{2}m(l^2 2\dot{\theta})\right] = m(2l\dot{l}\dot{\theta} + l^2\ddot{\theta})$$

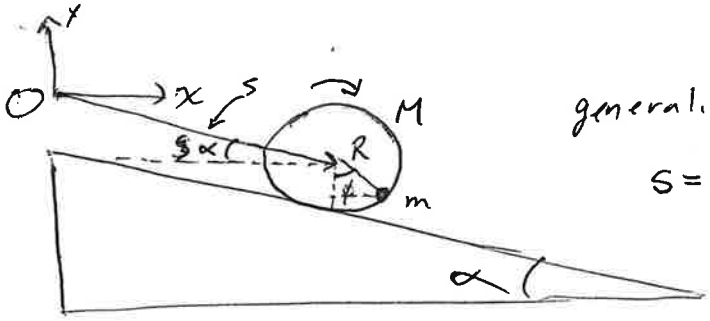
$$\Rightarrow \frac{d}{dt}\frac{\partial L}{\partial \dot{l}} - \frac{\partial L}{\partial l} = 0 \Rightarrow \frac{m}{m}\ddot{l} - \frac{m}{m}l\dot{\theta}^2 + \frac{k}{m}(l-b) - \frac{m}{m}g\cos\theta = \frac{0}{m}$$

$$\Rightarrow \boxed{\ddot{l} - l\dot{\theta}^2 + \frac{k}{m}(l-b) - g\cos\theta = 0}$$

$$\Rightarrow \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0 \Rightarrow m\left(\frac{2l\dot{l}\dot{\theta}}{l^2} + \frac{l^2\ddot{\theta}}{l^2}\right) + \frac{mgl}{l^2}\sin\theta = 0$$

$$\Rightarrow \boxed{\ddot{\theta} + \frac{2}{l}\dot{l}\dot{\theta} + \frac{g}{l}\sin\theta = 0}$$

5)



generalized coordinates $(s(t), \phi(t))$

$s = R\theta \rightarrow \dot{\theta} = \frac{\dot{s}}{R}$ → angular speed of disk

Disk (M)

$x = s\cos\alpha$

$y = -s\sin\alpha$

$\dot{x} = \dot{s}\cos\alpha$

$\dot{y} = -\dot{s}\sin\alpha$

Pendulum Bob (m)

$x_{bob} = s\cos\alpha + l\sin\phi$

$y_{bob} = -s\sin\alpha - l\cos\phi$

$\dot{x}_{bob} = \dot{s}\cos\alpha + l\dot{\phi}\cos\phi$

$\dot{y}_{bob} = -\dot{s}\sin\alpha + l\dot{\phi}\sin\phi$

$\dot{x}_b^2 + \dot{y}_b^2 = \dot{s}^2 + l^2\dot{\phi}^2 + 2l\dot{s}\dot{\phi}\cos\alpha\cos\phi - 2l\dot{s}\dot{\phi}\sin\alpha\sin\phi$

$$T = T_{disk} + T_{bob} = \left(\frac{1}{2}M\dot{s}^2 + \frac{1}{2}I\dot{\theta}^2\right) + \frac{1}{2}m\left[\dot{s}^2 + l^2\dot{\phi}^2 + 2l\dot{s}\dot{\phi}\cdot(\cos\alpha\cos\phi - \sin\alpha\sin\phi)\right]$$

$$\Rightarrow \boxed{T = \frac{1}{2}(M+m)\dot{s}^2 + \frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}ml^2\dot{\phi}^2 + ml\dot{\phi}\dot{s}\cos(\alpha+\phi)}\ \cos(\alpha+\phi)$$

$I_{disk} = \frac{1}{2}MR^2$, $\dot{\theta}^2 = \frac{\dot{s}^2}{R^2}$

$$\rightarrow \frac{1}{2}I\dot{\theta}^2 = \frac{1}{2}\left(\frac{1}{2}MR^2\right)\frac{\dot{s}^2}{R^2} = \frac{1}{4}M\dot{s}^2$$

$$V = V_{disk} + V_{bob} = -Mgs\sin\alpha + (mg(-s\sin\alpha - l\cos\phi))$$

$$\boxed{V = -(M+m)gs\sin\alpha - mgl\cos\phi}$$

$$L = T - V = \frac{1}{2}(M+m)\dot{s}^2 + \frac{1}{4}M\dot{s}^2 + \frac{1}{2}ml^2\dot{\phi}^2 + ml\dot{\phi}\dot{s}\cos(\alpha+\phi) + (M+m)gs\sin\alpha + mgl\cos\phi$$

$$\Rightarrow \boxed{L = \left(\frac{3}{4}M + \frac{m}{2}\right)\dot{s}^2 + \frac{1}{2}ml^2\dot{\phi}^2 + ml\dot{\phi}\dot{s}\cos(\alpha+\phi) + (M+m)gs\sin\alpha + mgl\cos\phi}$$

$L(s, \dot{s}, \phi, \dot{\phi})$

5) continued...

$$s: \frac{d}{dt}\frac{\partial L}{\partial \dot{s}} - \frac{\partial L}{\partial s} = 0, \quad \frac{\partial L}{\partial s} = (M+m)g\sin\alpha$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{s}} = \frac{d}{dt}\left[\dot{s}\left(\frac{3}{2}M+m\right) + ml\dot{\phi}\cos(\alpha+\phi)\right]$$

$$= \ddot{s}\left(\frac{3}{2}M+m\right) + ml\left(\ddot{\phi}\cos(\alpha+\phi) - \dot{\phi}^2\sin(\alpha+\phi)\right)$$

$$\Rightarrow \boxed{\left(\frac{3}{2}M+m\right)\ddot{s} + ml\left[\ddot{\phi}\cos(\alpha+\phi) - \dot{\phi}^2\sin(\alpha+\phi)\right] - (M+m)g\sin\alpha = 0}$$

$$\phi: \frac{d}{dt}\frac{\partial L}{\partial \dot{\phi}} - \frac{\partial L}{\partial \phi} = 0, \quad \frac{\partial L}{\partial \phi} = -ml\dot{\phi}\dot{s}\sin(\alpha+\phi) - mgl\sin\phi$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\phi}} = \frac{d}{dt}\left[ml^2\dot{\phi} + ml\dot{s}\cos(\alpha+\phi)\right]$$

$$= ml^2\ddot{\phi} + ml\left(\ddot{s}\cos(\alpha+\phi) \mp \dot{s}\sin(\alpha+\phi)\cdot\dot{\phi}\right)$$

$$\Rightarrow \cancel{m}\cancel{l^2}\frac{\ddot{\phi}}{\cancel{l^2}} + \cancel{m}\frac{l}{l^2}\left(\ddot{s}\cos(\alpha+\phi) - \cancel{\dot{s}\dot{\phi}\sin(\alpha+\phi)}\right) + \cancel{ml\dot{\phi}\dot{s}\sin(\alpha+\phi)} + \frac{\cancel{m}gl}{l^2}\sin\phi = \frac{0}{l^2}$$

$$\Rightarrow \boxed{\ddot{\phi} + \frac{1}{l}\ddot{s}\cos(\alpha+\phi) + \frac{g}{l}\sin\phi = 0}$$